

ENDFIELD DESKTOP DESIGN SYSTEM

A production-ready visual and interaction specification for a Linux desktop environment



Purpose: a self-contained system for creating new windows, tools, widgets, and desktop surfaces that remain visually and behaviorally coherent.

01 — Design intent

Create a focused, industrial-futurist desktop environment that feels engineered rather than decorative. The interface combines dark slate surfaces, high-visibility Endfield yellow, restrained blue/green status colors, technical metadata, and sharp 45° corner cuts.

The system must support a complete desktop: windows, dialogs, menus, panels, launchers, system widgets, notifications, settings pages, file tools, terminal tools, and third-party applications. Every new surface should be constructible from the primitives and rules in this document.

02 — Core principles

Clarity under density: information may be compact, but hierarchy, grouping, and state must remain obvious.

Functional ornament: serials, grid lines, brackets, and micro-labels should reinforce structure; never obscure the task.

Modular precision: use a small set of repeatable geometry, spacing, typography, and surface tokens.

State is visible: selected, focused, disabled, warning, syncing, and destructive states must be distinguishable without relying on color alone.

Quiet motion: transitions should feel like machinery engaging—short, linear-to-eased, and purposeful.

03 — Brand and visual language

Visual mood: field equipment, orbital infrastructure, tactical workstations, scientific instrumentation, and clean anime-industrial futurism.

Signature motifs: 45° clipped corners, yellow diagonal hazard bars, thin technical rules, compact uppercase metadata, outlined icons, and asymmetrical framing.

Do not use: glossy glassmorphism, soft consumer-app cards, excessive gradients, noisy HUD overlays, or uncontrolled neon.

04 — Color tokens

Use semantic token names rather than hard-coding colors. Themes may remap values, but semantic meaning must remain stable.

Token	Value	Usage
color.bg.900	#0B1118	Primary desktop / deepest surface
color.bg.800	#111A23	Window and panel background
color.bg.700	#1B2733	Raised surface / hover
color.line.500	#3A4A59	Borders and dividers
color.text.100	#E8EDF1	Primary text

color.text.300	#AAB7C2	Secondary text
color.accent.500	#FFD400	Brand accent / focus / attention
color.info.500	#4B9BFF	Informational state
color.success.500	#8BD32E	Success / connected
color.danger.500	#FF6B45	Critical / destructive

05 — Typography

Primary UI typeface: Inter, IBM Plex Sans, or an equivalent highly legible sans-serif. Use a condensed technical face only for metadata, serials, and display labels.

Typography is mostly left-aligned. Uppercase is reserved for navigation labels, metadata, status tags, and section headings. Avoid all-caps for long body copy.

Style	Size / line	Weight	Use
Display / H1	32 px / 36 px	700	Page title, major desktop mode
H2	20 px / 24 px	650	Window and section title
H3	14 px / 18 px	650	Subsection, card title
Body	13 px / 20 px	400	Default UI copy
Label	11 px / 14 px	600	Control labels, tabs
Micro	9 px / 12 px	500	Serials, metadata, technical notes
Numeric	24 px / 28 px	650	Primary metric / clock

06 — Geometry, spacing, and depth

Base unit: 4 px. Preferred spacing values: 4, 8, 12, 16, 20, 24, 32, 40, 48 px.

Default corner treatment: 45° clipped outer corners. The cut is a geometric replacement for a radius, not an additional decoration.

Default panel border: 1 px. Emphasis border: 2 px. Standard shadow is low-opacity and offset downward; avoid floating layers that look detached from the desktop.

Use a 12-column layout for full-screen surfaces and an 8 px baseline grid for internal alignment.

07 — Surface and layering model

Layer 0: wallpaper / environment image.

Layer 1: desktop chrome—top bar, dock, desktop widgets.

Layer 2: application windows and utility panels.

Layer 3: menus, popovers, context menus, tooltips, command palettes.

Layer 4: modal dialogs, critical alerts, authentication, and system lock surfaces.

Each layer receives stronger contrast and a higher scrim only when needed. Never stack multiple opaque black rectangles without hierarchy.

08 — Window system specification

Every window uses the same anatomy: title strip, window identity, optional serial label, window controls, content region, optional status strip, and clipped-corner frame.

Window modes: standard, utility, modal, full-screen, pinned, minimized, and transient popover.

Title strip: 36–44 px high. Left side contains app icon, title, and optional context. Right side contains minimize, maximize/restore, and close controls.

Active window: yellow focus rail or yellow corner marker. Inactive window: lower contrast title and border, but retain readable controls.

Resize affordance: 8 px hit area at edges/corners; visual affordance is subtle and appears on hover.

Minimum window size: 320 × 240 px unless the application defines a stricter minimum.

09 — Component inventory

Required primitives: clipped panel, technical divider, icon button, text button, segmented control, tab, input, search field, dropdown, checkbox, switch, slider, progress bar, badge, status dot, tooltip, toast, table row, list row, tree row, breadcrumb, avatar/operator marker, empty state, skeleton loader, modal, drawer, context menu, and command palette.

Required composites: application header, sidebar navigation, toolbar, data table, inspector panel, file grid, terminal pane, settings section, notification item, quick-setting tile, workspace switcher, media control, and system metric card.

Component	Construction	Rule
Window frame	Surface + 1 px line + 45° cuts + title strip + active rail	Use for app windows and major utility surfaces.
Icon button	24–36 px hit area, 16–20 px icon, clipped corner on hover	Always provide tooltip and accessible name.
Technical badge	Label + semantic dot/icon + optional serial	Use for connection, sync, mode, or priority.
Toolbar	Horizontal flex row, 8 px gaps, divider groups	Keep primary actions left; overflow secondary actions.
Data row	32–40 px height, aligned columns, selected rail	Use for tables, logs, processes, and inventory.

10 — Desktop chrome

Top bar: left app/menu trigger; center or left workspace context; right system indicators, connectivity, battery/power, clock, and profile/session status.

Dock: centered or edge-aligned; icons sit on a dark slate surface with a yellow active marker. Icons are 20–24 px, with 8 px gaps and a 4 px active underline or corner marker.

Desktop widgets align to the same grid as windows. Default widgets: clock/date, system telemetry, weather/environment, mission/task list, network status, storage, and quick actions.

11 — Widget construction rules

Every widget must define: name, purpose, data source, refresh cadence, size variants, loading state, empty state, error state, interaction model, keyboard behavior, and accessibility label.

Widget anatomy: header (label + status), primary value or content, secondary metadata, optional action, and a technical footer/serial.

Small widgets prioritize one action or one metric. Medium widgets may contain a list or chart. Large widgets may contain tabs, filters, and multiple data groups.

Charts use thin lines, small data points, restrained fills, and semantic colors. Never rely on yellow alone to communicate danger.

12 — Interaction and motion

Hover: 80–120 ms opacity or border response.

Press: 60–100 ms scale/contrast response; avoid exaggerated bounce.

Open/close: 160–220 ms. Use ease-out for entering and ease-in for leaving.

Panel transitions: translate 4–12 px plus opacity. Avoid large zooms.

Focus must be persistent and keyboard-visible. Motion can be reduced through a system preference.

13 — States and semantics

Neutral: slate/gray border and text.

Informational: blue.

Success/connected: green.

Warning/attention: Endfield yellow.

Critical/destructive: red-orange.

Disabled: lower contrast and no active decoration, while maintaining readable contrast.

Pair every semantic color with an icon, label, shape, or position cue.

14 — Iconography

Use a consistent 1.5–2 px stroke icon set with squared or slightly chamfered terminals. Icons should be geometric, compact, and recognizable at 16, 20, and 24 px.

Do not mix filled cartoon icons with technical outline icons. Brand marks may be exceptions.

Icon buttons require accessible names and a tooltip after a short hover delay.

15 — Data density and technical metadata

Micro-data labels use 9–11 px type, increased tracking, and muted slate/blue-gray color. Examples: EF-2037-1186, SYS/NET/04, SESSION 08:42, NODE ACTIVE.

Metadata is supportive, not primary. Use it to establish authenticity and system context, never to replace a meaningful label.

Tables should favor alignment, compact rows, visible column labels, and clear selected-row treatment.

16 — Accessibility and resilience

Target WCAG AA contrast for text and essential controls. Do not use the background image as the only contrast surface behind text.

Keyboard order follows visual order. Every interactive element needs a visible focus state, name, role, and state.

Support 100–200% text scaling, reduced motion, high-contrast mode, light theme fallback, and color-blind-safe semantic pairings.

Provide tooltips or expanded labels for icon-only controls. Error messages must explain what happened and how to recover.

17 — Implementation checklist

Before shipping a new window or widget, verify: token usage; 4 px spacing grid; clipped-corner geometry; active/inactive states; keyboard navigation; focus visibility; loading/empty/error states; responsive minimum size; semantic color pairing; serial/metadata restraint; reduced-motion behavior; and light/high-contrast fallback.